

Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

TEAM ID	SWTID-2026-6798
PROJECT TITLE	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
MAX MARKS	5 MARKS

CHAPTER – IV PROJECT DESIGN

4.1 – Problem Solution Fit

The primary problem addressed in this case study is the growing concern over unhealthy eating habits among college students, which can negatively affect their physical health, academic performance, and overall well-being. Many students face challenges such as limited budgets, busy schedules, lack of nutritional awareness, and easy access to fast food, leading to poor dietary choices. Traditional methods of monitoring student nutrition often fail to provide comprehensive insights into eating behaviors and the factors influencing them. To address this challenge, the proposed solution leverages a modern technology stack combined with Tableau's powerful data visualization capabilities to collect, process, analyze, and present food consumption data in an interactive and meaningful way. Data from student surveys, cafeteria transactions, nutritional databases, and demographic records are integrated into a centralized system, cleaned through data-processing tools, and visualized using Tableau dashboards. These dashboards reveal patterns in meal consumption, nutritional intake, spending behavior, and food preferences across different student groups. The solution enables educational institutions, nutritionists, and campus administrators to identify dietary deficiencies, recognize at-risk student populations, and make evidence-based decisions. Based on the insights generated, targeted dietary strategies such as personalized meal recommendations, healthier cafeteria menus, nutrition awareness programs, and budget-friendly healthy meal options can be implemented. The problem–solution fit is strong

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because the analytical framework directly addresses the root causes of poor dietary habits while providing actionable insights for continuous improvement. By transforming raw food-related data into clear visual intelligence, Tableau helps stakeholders monitor trends, measure the effectiveness of interventions, and promote healthier food choices across the campus community. Ultimately, this data-driven approach enhances student health outcomes, supports informed decision-making, and creates a sustainable foundation for improving nutrition and wellness in higher education institutions.

